Life Insurance and Wealth: What Shapes Participation and Do Tax Incentives Matter?

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Abstract

Welfare states across the OECD incentivize life insurance take-up. Using a novel administrative data set on wealth at death from Vienna, Austria, we study how death benefits from life insurances supplement the balance sheets of individuals who die with low wealth, and how they respond to an income tax reform that changed the tax treatment of insurance premia. At least 38.7% of estates comprise little or no wealth. Death benefits account for 18% of terminal assets in the bottom five deciles of the distribution of estates, on average. Most payouts are final-expense policies that insure funerals. Take-up of final-expense insurance responds to tax incentives, while other life insurance and net wealth at death do not. Our results shed new light on the motives to take out life insurance, the distributional importance of benefits and the tax treatment of insurance.

JEL Codes: D1, D3, G5, H24, H26, J14

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1 Introduction

Life insurance constitutes one of the largest insurance markets, especially in advanced economies (OECD 2024). As a source of income for the dependents of the policy holder upon death, and as a long term savings vehicle that can accumulate a cash value, life insurance is a tax-subsidized element of the old-age safety net in welfare states across the OECD (OECD 2023; OECD 2005; Horn and Kohl 2024; Gerba and Schelkle 2013; Palier 2007). Yet, are holders of life-insurance policies better off than others in terms of late-life wealth? And to what extent is this advantage a consequence of preferential tax treatment?

Although it is optimal for consumers to pass away with zero net wealth under certain conditions (Modigliani and Brumberg 1954), there are reasons to be concerned about extremely low or negative end-of-life wealth. First, the lack of terminal wealth can generate externalities if costs are borne by survivors, governments, and other creditors who absorb late-life expenditure shocks, cover unpaid obligations, or forego claims on the estate. Second, precarious individual finances at death are an indicator of more persistent disadvantage, and wealth in late life or at death is a good proxy for wealth among the living (Alvaredo, Atkinson, and Morelli 2018; Poterba, Venti, and Wise 2018).

Against this backdrop, this paper proceeds in three steps, drawing on a novel dataset of digitized probate court files from Vienna, Austria. We begin by showing that low wealth at the end of life is associated with economic insecurity for many decedents. As life insurance (and hence death benefits) is one of the most important assets for low-wealth individuals, we next analyze which individual characteristics are associated with insurance take-up. Finally, we exploit a tax reform that temporarily increased the relative attractiveness of insurance participation to study whether tax incentives can raise insurance take-up and, in turn, boost wealth at the bottom of the distribution.

We present several novel results. First, in addition to providing evidence on the economic situation of low-wealth decedents, our data reveals that death benefits account for one fifth to a quarter of assets at death in the lower half of the distribution. Second, we show that the majority of death benefits insure funerals, rather than the consumption needs of survivors, as decedents in our data tie up death benefits in final expense insurance policies.[1] We find that demand for final expense insurance is not driven by the absence of medical underwriting. Other types of death benefits are more closely linked to the presence of survivors and show a clear hump-shape in the decedent's age. Third, our evaluation of the tax reform suggests that take-up of final expense insurance is sensitive to tax incentives, while other life insurance and net wealth at death are not. This suggests that people simply shift their portfolios in response to taxation, rather than increasing wealth accumulation.

Our perspective on wealth at death is insightful for the study of economic well-being and death benefits. First, our administrative data complements evidence from survey data, which has merits but also important drawbacks for the study of wealth, including issues related to coverage and frequency.[2] Second, the rich information in our data permits a distinction between death benefits from different types of life insurance. This granularity allows us to establish novel facts about insurance and take-up motives. Third, most evidence on life insurance comes from surveys of the living. But many

[1] Death benefits of final expense insurances can only be spent on funerals and related costs and policies are available already at low amounts of cover and without medical exams ("Medical underwriting").

[2] In contrast to household surveys that exclude institutional households (Poterba, Venti, and Wise 2018), struggle to cover the rich (Disslbacher et al. 2023) and often lack individual-level wealth data, our dataset includes individuals in institutions, over-samples the rich and refers to individuals. As opposed to most administrative data sources that cover specific segments of the distribution (Berman and Morelli 2021), the data generating process underlying our study applies to everybody who passes away.

policies only pay if the insured dies while coverage is in force. If insurance coverage among decedents differs systematically from coverage among the living (Cawley and Philipson 1999), the benefits actually paid at death can diverge from the coverage one would infer from a cross section of living individuals.[3] Studying realized death benefits reveals the true distributional impact of life insurance on decedents and those who receive bequests.

This paper contributes to the literature on household finance and insurance (Gomes, Haliassos, and Ramadorai 2021; Hong and Ríos-Rull 2012; Bernheim 1991) by examining life insurance, and providing a more granular distinction between final expense insurance (FI) and other types of life insurance (LI).[4] Our results are consistent with recent findings on insurance participation and reveal several important new facts about life insurance. First, we show a hump-shaped relationship between final expense insurance take-up and wealth rank. For other life insurance products, the decline in participation at higher wealth levels is much less pronounced, such that the relationship between wealth and take-up is positive (Gropper and Kuhnen 2025). Second, in terms of demographic characteristics, the results reveal that final expense insurance is popular among women. Marital status is much less predictive of final expense insurance than it is of other types of life insurance, which has previously been found to be more heavily used by married

[3]One explanation is asymmetric information between insurers and consumers. Cawley and Philipson (1999) find evidence that insurers may assess mortality risk more accurately than consumers. In the extreme, if insurers' information were perfect, minimal coverage among decedents could coexist with high coverage among the living. There might also be a mismatch between expected benefits and actual payouts, when individuals struggle to understand complex benefit rules or promised benefits do not match those provided (Hacker et al. 2018).

[4]While funeral insurance has been discussed in the context of developing countries (Banerjee et al. 2024), there is little comparable evidence for advanced economies. There have been specific attempts to model final expense insurance in the developing-country context, where this type of policy is used as an intergenerational commitment device (Berg 2018). Interestingly, in our paper we find that survivors' funeral expenditure does not depend on whether the decedent is insured or not. Appendix G provides details.

individuals (Inkmann and Michaelides 2012). The same holds for age.[5] Third, the patterns of insurance holdings suggest that the presence of medical underwriting might not be a key factor in shaping demand for certain insurance products.

Moreover, the paper sheds new light on economic well-being. Inspired by the literature on asset-poverty that studies whether individuals can draw on private wealth to weather specific contingencies (Rapp and Humer 2023; Kuypers et al. 2022; Brandolini, Magri, and Smeeding 2010; Haveman and Wolff 2004),[6] we find that 38.7% of decedents left estates valued at less than three months of consumption at the Austrian income poverty line (€ 1,322.67 per month for a single household) before subtracting final expenses. Not all of these individuals were economically vulnerable in late life, defined as unprotected against hardship-causing economic losses. However, our calculations expose some degree of vulnerability, showing that 45% of low-wealth individuals have either significant debt, low incomes or received means-tested transfers before death. A known weakness of research in this area is the limited attention it pays to formal insurance, as opposed to self-insurance (Hacker et al. 2018). We quantify the importance of private insurance along the distribution of wealth,[7] and show that it complements self-insurance especially in the lower middle of the distribution.

Our study also adds to the literature on taxation. First, changes in the taxation of life insurance premia have been studied before in Germany and Italy. The two German

[5] Research on life insurance and age is more inconclusive. Some papers find a positive relationship between insurance usage and age, while others document a decline with age (Gomes, Haliassos, and Ramadorai 2021). Bulmahn (2003) and Lin and Grace (2007) find that life insurance take-up follows an inverted U-shape in the US and Germany.

[6] In a related literature on the measurement of material deprivation, financial buffers also play an important role: Somebody's ability to cover sudden financial expenditures such as a funeral, a medical bill, or the replacement of a durable consumer good is a key element of material deprivation in the official Eurostat definition (Marlier et al. 2007; Gilbert 2009) and beyond (Guio et al. 2016).

[7] Several papers show how the public safety net helps people to cope with high out-of-pocket expenditure associated with death (Valentine and Woodthorpe 2014; Woodthorpe, Rumble, and Valentine 2013), this paper is the first to illustrate the role of private final expense insurance in achieving this goal.

studies suggest that tax incentives shift insurance take-up (Hecht and Hanewald 2012; Sauter and Winter 2010). Jappelli and Pistaferri (2003) do not find an effect of the Italian reform on household portfolios. We do not only complement these papers by using administrative rather than survey data on wealth, but also by showing that the responses to the reform differs across insurance products. Second, there is a literature on taxation and wealth at death. While prior work has focused on inheritance taxation and its effects on terminal wealth (Suari-Andreu et al. 2024; Escobar, Ohlsson, and Selin 2023; Glogowsky 2021; Erixson and Escobar 2020; Kopczuk 2007; Brunetti 2006; Poterba and Weisbenner 2003; Slemrod and Kopczuk 2000), we instead study how income tax policy shapes wealth at the end of life. The literature on income taxation and wealth at death is more limited, and evolves around specific elements of the interaction between income and estate taxation, such as the Basis Step-Up at Death (Gordon, Joulfaian, and Poterba 2016; Kopczuk 2016) or around means testing of income transfers and wealth (Johannesen, Sæverud, and Saez 2024; Wellschmied 2021; Neumark and Powers 1998). We document some degree of responsiveness to tax incentives especially among less affluent households. These individuals are often not covered in data from estate or inheritance tax returns, because taxation usually only affects more affluent decedents.

In terms of policy-making, low wealth among the elderly is a serious concern (Poterba, Venti, and Wise 2018; Gornick, Sierminska, and Smeeding 2009). Beyond the direct consequences for individuals facing economic vulnerability or even material deprivation, insufficient financial buffers can cause expenditure shocks to propagate through private family support networks and, ultimately, the welfare state. Encouraging private wealth accumulation and greater life insurance participation may help address these challenges (Harris and Yelowitz 2018; Weir and Willis 2000; Hurd and Wise 1989). While our results show that death benefits perform an important buffering role to end-of-life expenditure shocks, they also suggest a cautionary tale regarding tax incentives as an instrument to reduce economic vulnerability. We reach this conclusion because tax advantages do not translate into more wealth at the point where the insurance becomes operative. As a result, it may be preferable to consider alternative means to reduce financial vulnerability.

The remainder of the paper is organized as follows. In Section 2, we set out in detail the characteristics of final expense insurance and how it contrasts with other types of insurance. Subsequently, we introduce our data set (Section 3) before presenting the findings in Section 4. Section 5 concludes.

2 Insurance products and institutional environment

Life insurance pays a benefit upon the death of the insured individual. A broad range of products exists. Among insurance products, we distinguish term life insurance products and whole life insurance that accumulates a cash value (Hong and Ríos-Rull 2012). The first family of insurances provides a death benefit over the coverage period and no benefit if the insured individual survives the coverage period. Whole life insurance has a lifelong coverage period. As a result, individuals receive the face value with certainty at some point, and therefore they pay higher insurance premia.

There are several additional important characteristics that introduce variation between life insurance policies. First, some life insurance contracts are only available at a minimum/maximum level of cover (and hence premium). Second, life insurance products differ in terms of the fungibility of the benefit payment. Most importantly, a large share of insurance payouts at death are earmarked for final expenses. Finally, some insurance products have medical underwriting. Medical underwriting implies that insurance products might either not be available for individuals in poor medical condition, or that the insurance provides might charge higher costs.

Against this backdrop, we distinguish broadly between two types of life insurance: products marketed as final expense insurance (FI) and other types of life insurance (LI). Typical final expense insurance policies in Austria are similar to a whole life insurance with a low face value. Historically, FI played an important role in popularizing life insurance in the broader population, facilitated by low premia and small face values (Hadziabdic and Kohl 2022). In most cases, the insurance is a limited pay life insurance. In this arrangement, the policy holder pays their premium only for a set time period rather than their whole life, while maintaining lifelong coverage. Table 8 in Appendix B provides an overview of several funeral insurance contracts available in Austria in 2017. In addition, FI are usually characterized by benefits that are at least to some extent earmarked for final expenses and feature minimal medical underwriting. Insurances that are not marketed as FI fall into the LI category. Table 1 summarizes the key differences between both types of life insurance.

Table 1: Funeral insurance vs. other life insurance

Variable	**Funeral insurance (FI)**	**Other life insurance (LI)**
Earmark	✗	✓
Medical underwriting	✗	✓
Minimum cover	$\downarrow$	$\uparrow$
Maximum cover	$\downarrow$	$\uparrow$

Notes: $\downarrow$ = low, $\uparrow$ = high; ✗ = no, ✓ = yes.

Until 2016, premium payments to funeral insurances benefit from tax deductibility. The tax deduction is capped at € 2,920 for childless[8] individuals with a taxable income

[8] Individuals with children benefit from more generous deductibility. Single parents had a maximum deduction of EUR 5,840, while for parents with three children or more, the deduction increased by an additional EUR 1,640.

below 36.400, and gradually falls above that income. The maximum income to claim the tax deduction is €60,000.

Until 2016, premium payments to funeral insurances benefited from tax deductibility ("Topf-Sonderausgaben"). The tax deduction was capped at EUR 2,920 for childless individuals with a taxable income below EUR 36,400, and gradually falls above that income. The maximum income to claim the tax deduction was EUR 60,000. Final expense insurance premia were not the only expense that taxpayers could deduct from their tax liability. Premia for other personal insurances (voluntary health and accident insurances as well as certain term life insurances) as well as certain costs connected to the creation of residential homes were also deductible.

On March 13th 2015, the federal government of Austria announced a tax reform, scheduled to take effect on the first of January, 2016. The reform phased out the tax deductions for personal insurances. Specifically, insurance premia payments for contracts concluded before the reform became effective continue to be tax deductible until 2020. In contrast, individuals who have taken out insurance since the reform was operative are no longer able to reduce their tax liability by deducing insurance payments (even if they are made between during the transitional period between 2016 and 2020). We discuss other key features of the tax reform in the Appendix E.

Overall, the reform incentivizes taxpayers to take out an insurance contract before 2016, but after the reform was announced. The reform increased the price of taking out insurance only if insurance contracts were concluded after the transitional period already started. For existing contracts and those that were made before 2016, the transitional period gave people the chance to deduct a large share of their final expense insurance premia. This is because many insurance companies offer flexible payment schemes, and allow consumers to pay all premia in several large payments or even a single one-off payment.

3 Data

We compile a dataset that allows a joint analysis of wealth at death, the portfolios of decedents, their choices regarding wealth transfers to survivors as well as a broad range of socio-demographic characteristics. The data is based on a sample of digitized probate records.[9] Probate records are court files that document each step in every probate court case in Austria. The purpose of this procedure is to document all assets and liabilities of each decedent exhaustively in order to enforce the inheritance law and transfer ownership titles to heirs. In addition, the probate process ensures that all entries in official registers are made ascertain that other final arrangements are made. Probate data has an important role in research on wealth at death (Tomes 1981; Menchik and David 1983; Brunetti 2006). In contrast to other countries such as the UK, by Austrian law a probate proceeding is initiated for every death, irrespective of the level or composition of assets held by the deceased. This results in a wide population coverage.

Our sample of probate records is drawn from probate courts in 10 districts of Vienna, the capital, covering the years from 2014 to 2019. In each district, we sample approximately 11% of probate cases each year. After dropping probate cases that were processed by foreign jurisdictions yields a total number of 4712 observations.[10] We predominantly use a stratified approach to draw the sample, to ensure that the data covers extreme values. To achieve an oversampling of high-wealth probate cases, the strat-

[9] We compare our data to other data sources on wealth in Vienna using a matching approach in a companion paper.

[10] We also drop homeless individuals. All probate cases of individuals who pass away without an address of permanent residence are processed in the the district court "Innere Stadt". Including them in the sample would lead to a strong overrepresentation of the homeless.

ification of the selection within the court districts aims to draw particularly complex proceedings with a higher probability. In nine out of the ten districts in our sample, we replace 5% of the randomly drawn sample within each district-year with the probate cases that have the most procedural steps in that district-year. This approach is based on the assumption that complex proceedings with more procedural steps are also associated with higher estate values.[11]

The probate data contain information on all assets and liabilities valued at the point of death of the decedent. The individual balance sheets cover real estate, vehicles, business wealth, valuables, financial wealth, cash, and claims against other individuals or organizations. In contrast to the Austrian Household Finance and Consumption Survey (HFCS) data, wealth is at the individual level, and jointly held assets are allocated to the decedent by the probate court based on their share or on an equal split basis if assets are jointly owned. The broad coverage of different assets is an important characteristic that distinguishes our dataset from other probate records. For example, as opposed to the English probate data, jointly held property is included in this paper's wealth measure. Moreover, in contrast to the English data, it is possible to extract information on portfolios and specific wealth components from our dataset, such as housing wealth.

Liabilities include bank loans, credit overdraft, unpaid bills, and obligations towards other individuals. We do not add liabilities that only materialize at death to the measure of debt. This includes funeral costs, probate court and notary fees, but also the asset recovery claims from public minimum income support transfers, for example.

Survivors have strong incentives to report assets truthfully as misreporting is under threat of punishment. For most financial assets, a proof of the value, such as a bank

[11] In the Appendix, we show that the number of procedural steps is a correlate of the duration of a probate case. It is noteworthy that our over-sampling approach is an improvement over the Austrian Household Finance and Consumption Survey, which does not engage in oversampling.

statement is required. In many cases, professional valuators are involved, not least for real estate. While under certain circumstances, gifts made in the years prior to death are included in the probate process, this is by far not the case for all inter-vivos transfers. Therefore, we cannot add the value of inter-vivos transfers to the estate.

In addition to information on assets and liabilities, we extract data on demographic characteristics, including marital status, gender, age and tenure status. Furthermore, the data contains limited information on incomes. For example, we observe regular transfers from the social security administration for some decedents, which we consider as retirement income. Whenever we use income data in this paper, we restrict the sample to observations where we have this information (25% of the sample).

The dataset permits unique insights into final expense insurances compared to other sources. Other data sources such as the HFCS or the Survey on Health and Retirement (SHARE) do not contain this information. Other data sources do not exist: the abolishment of wealth and inheritance taxation and the specific rules for capital income taxation make it impossible to use tax data for studying individual financial behavior in Austria. Figure 4 in the Appendix A.2 tracks the aggregate evolution of different types of life insurance implied by our data for Austria. Figure 5 compares the aggregate value of life insurance in our data to the aggregates from other sources such as the financial accounts.

4 Results

This section first examines in more detail the economic situation of decedents with low wealth, and the importance of death benefits along the distribution of wealth at death. Next, Subsection 4.2 examines the correlates of death benefits descriptively, focusing on socio-demographic characteristics of decedents. We compare FI and LI death benefits. Subsection 4.3 estimates the responses of decedents to the 2016 income tax reform in terms of their final expense planning behavior.

4.1 Financial buffers among decedents

A substantial share of decedents left behind very limited wealth. Using the income poverty line for a single-person household in Austria (€ 1,322.67 per month) as a benchmark, 38.7 percent of estates contained less than the equivalent of three months of consumption at the poverty line.

Is low wealth at death an indicator of poor financial well-being? Figure 1 gives a tentative answer to this question. It sheds light on other indicators of low financial well-being among individuals with low wealth. Low wealth, low income, high debt, and involvement in an asset recovery (AR) process each capture different aspects of financial distress at the end of life. Low wealth is defined as terminal net wealth below one quarter of the annual income at the poverty line for a single individual, while high debt refers to net wealth below this threshold multiplied by negative one. Low income denotes individuals in the bottom quintile of the income distribution within our sample. Figure 1 highlights that these forms of economic disadvantage frequently coincide. However, a significant share of decedents with low wealth can rely on their income to support consumption, do not have any significant debt and also do not depend on means-tested transfers during their life-time.

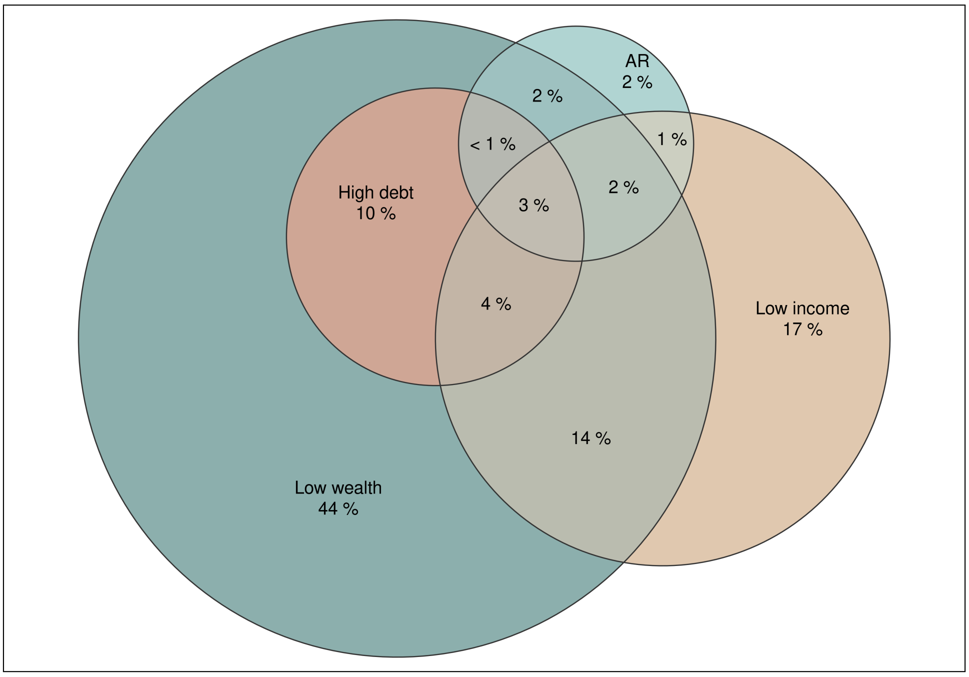
Among all decedents with valid income data and at least one form of low financial well-being, 44% exhibit low wealth without simultaneously having low income, high debt, or involvement in an AR process. However, a significant minority of those who pass away with low wealth (45% of the largest dark blue circle) have low economic well-being in at least one other dimension. It is also notable that 39% of individuals who pass away with assets below a quarter of the annual income poverty line for single households have significant debt. Approximately three percent of decedents have low assets and high debt, in addition to low incomes and AR. These patterns indicate that low terminal wealth is not simply a reflection of life-cycle behavior or intentional wealth decumulation in anticipation of death. For a significant share of decedents, low wealth coincides with other forms of economic insecurity, pointing towards more persistent financial fragility.

Figure 2 summarizes the portfolio composition across deciles of gross and net wealth at death. Death-benefit products (LI and especially FI) are prominent in the lower half and the middle of both distributions. In the fifth gross-wealth decile (upper panel), death benefits account for roughly one quarter of assets, about two thirds of which is FI. Their importance peaks in the fifth decile, remains sizable in the sixth, and then declines sharply toward the top. The net wealth distribution (lower panel) shows a similar pattern. Unlike the gross-wealth case, however, individuals in the second and third net-wealth deciles also hold substantial insurance. This implies that decedents who have low net wealth due to debt hold insurance. On average, in the bottom five deciles of the net wealth distribution, LI and FI account for 18% of terminal assets.

4.2 Determinants of death benefit payments

Table 2 examines the correlates of funeral insurance and other life insurance, both in the extensive margin in columns (1) and (3), as well as in the intensive margin in column (2) and (4). The first and third column refer to the marginal effects estimated with a

Figure 1: Economic well-being of low-wealth decedents

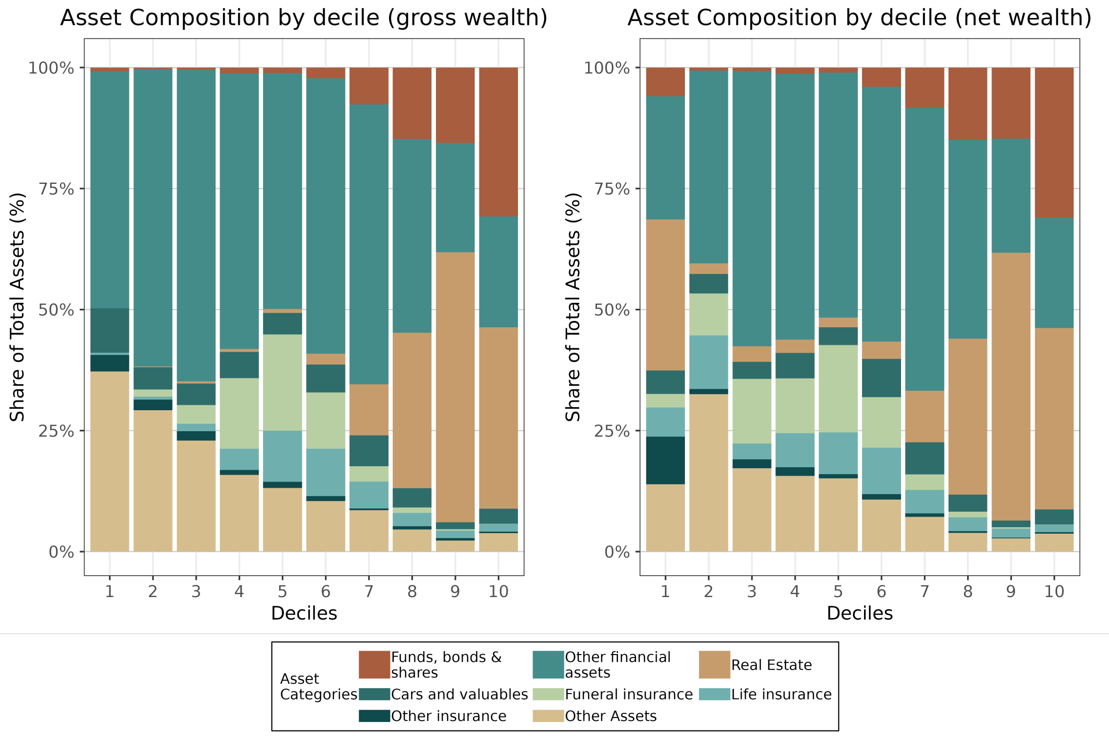


Note: This Euler diagram illustrates the overlap between several dimensions of economic well-being among individuals who, at death, have low net wealth, low income, or estates subject to asset recovery by the local government (AR), or who exhibit a combination of these characteristics. The sample includes only decedents with valid income data. Low wealth is defined as terminal net wealth below one quarter of the annual income at the poverty line for a single individual. High debt is defined as net wealth below the same threshold multiplied by negative one. Low income refers to individuals in the lowest quintile of the income distribution within our sample. The percentages indicate the share of all decedents exhibiting at least one of these four characteristics. For instance, 44% of decedents with low wealth do not also have low income, high debt, or estates involved in an asset recovery process. Weights are used to take into account stratified sampling.

probit model. Insurance take-up as the dependent variable. The results on the intensive margin in the second and fourth column are based on a model that corrects for selection into the sample of insured individuals (Heckman 1979). All specifications feature court district and year-of-death fixed effects.

The comparison of life and funeral insurance policy holdings at the end of life shows several interesting facts about the relationship between demographics and insurance

Figure 2: Portfolios along the distribution of terminal wealth.



Note: The left panel refers to the share of different types of assets in total gross wealth for the deciles of the gross wealth distribution of decedents. The right panel refers to the share of different asset types in total assets for the deciles of the net wealth distribution of decedents. "Other financial assets" refer to cash, checking and savings bank accounts and building society savings contracts. "Other assets" includes claims towards other individuals, companies or the government. Weights are used to take into account stratified sampling. Results are pooled over years.

participation. First, the age-dependence of funeral insurance is not as clear as the age-dependence of other life insurance products. while older individuals tend to have significantly higher levels of payouts at death in column (2), they are not more likely to take out an insurance policy. In contrast, life insurance participation is related to age in an inverted u-shape, both in terms of the intensive and the extensive margin. Second, Table 2 reveals parallels between different insurance products. Both funeral insurance but also life insurance more generally are in higher demand among widowed individuals, and women. Women also participate more actively in funeral insurance on the intensive margin, compared to men. While single decedents are neither more nor less likely to participate in funeral insurance, singles are significantly more likely to have some other sort of life insurance. Using an indicator for long-term care allowance recipients as a proxy for individuals in poor health, we do not find systematic variation across health states in view of insurance demand.

Table 2 also reflects how other financial choices that individuals make towards the end of their live relate to funeral insurance cover. Neither residential choice nor inter-vivos gifting is associated with higher insurance participation. However, there is robust evidence showing that decedents with (higher levels of) funeral insurance are less likely to die intestate. This finding does not hold for other types of life insurance. Finally, in terms of net wealth at death, there is a pronounced hump-shaped relationship to funeral insurance participation. The hump-shape is particularly significant for the extensive margin, with weakly statistically significant coefficients for the intensive margin. A similar relationship governs the take-up of other life insurance products, although the inverted u-shape is much more pronounced for funeral insurance. Participation for life insurance also peaks at a lower wealth rank (see also Figure 6).

4.3 Life insurance and tax policy

Are life insurance holdings sensitive to tax incentives? To investigate this question based on the 2016 tax reform in Austria, we examine participation among decedents in granular time intervals around the reform announcement. Figure 3 plots the share of insured decedents in month-of-death bins from 2014 through the end of 2019. The red vertical line indicates the announcement of the tax reform in 2015. The shading and size of the points indicates the number of decedents within each bin. For ease of comparison,

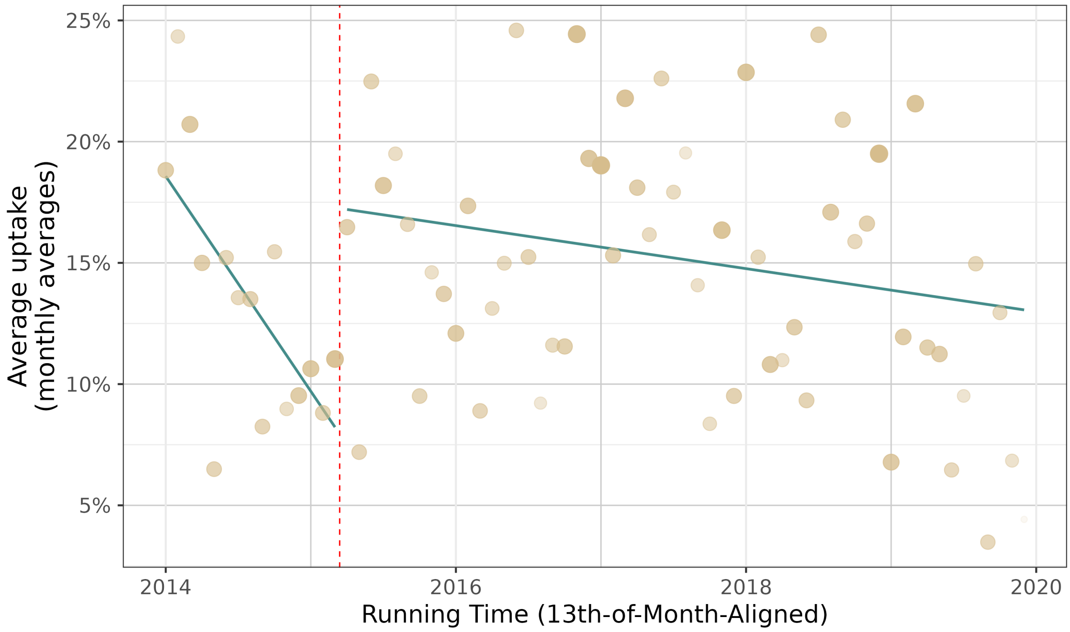
Table 2: Correlates of final expense and life insurance uptake

	Dependent variable: uptake / amount			
	probit FI (extensive)	*OLS* FI (intensive)	*probit* LI (extensive)	*OLS* LI (intensive)
	(1)	(2)	(3)	(4)
Age (linear)	0.002	0.011	0.004^{***}	0.133^{**}
	(0.003)	(0.008)	(0.001)	(0.052)
Age (quadratic)	0.00001	−0.00004	-0.0001^{***}	-0.002^{**}
	(0.00002)	(0.00003)	(0.00001)	(0.001)
Marital status: Married	−0.017	−0.050	−0.006	0.419^{*}
	(0.019)	(0.107)	(0.017)	(0.254)
Marital status: Widowed	0.026	−0.010	0.021	1.086^{***}
	(0.021)	(0.104)	(0.019)	(0.365)
Marital status: Divorced	0.016	0.029	0.011	0.721^{***}
	(0.023)	(0.087)	(0.019)	(0.266)
Female	0.024^{**}	0.171^{*}	0.017	0.405
	(0.012)	(0.099)	(0.011)	(0.260)
With care allowance	0.018	−0.024	0.001	−0.210
	(0.016)	(0.079)	(0.014)	(0.199)
Has a testament	0.028^{**}	0.189	−0.009	−0.210
	(0.013)	(0.120)	(0.012)	(0.202)
Made a gift	−0.027	−0.160	0.009	0.334
	(0.022)	(0.186)	(0.023)	(0.211)
Net Wealth (perc.) (linear)	0.009^{***}	0.046	0.006^{***}	0.186^{**}
	(0.001)	(0.032)	(0.001)	(0.081)
Net Wealth (perc.) (quadratic)	-0.0001^{***}	−0.0004	-0.00004^{***}	-0.001^{**}
	(0.00001)	(0.0003)	(0.00001)	(0.001)
Has LI	0.146^{***}	0.619		
	(0.015)	(0.516)		
Has FI			0.124^{***}	2.294
			(0.012)	(1.716)
Observations	4,172	618	4,172	501

Note:: Dependent variable is an indicator for insurance uptake (extensive) and the log payout sum at death for decedents with positive payouts (intensive). FI (LI) is final expense (other life) insurance. Results from probit regressions are reported as marginal effects. Robust (HC0) standard errors in parentheses; regressions weighted to take into account stratified sampling. All models feature year-of-death fixed effects and court district fixed effects. Fixed effects and coefficients on residual categories for categorical variables dropped.
*p<0.1; **p<0.05; ***p<0.01

we overlay separate linear fits before and after the announcement to highlight levels and trends.

Figure 3: Discontinuity in insurance coverage at death before and after reform



Note: Insurance participation in one-month time intervals before and after the announcement of the 2016 tax reform. Each point represents the share of insured decedents in the total number of individuals who passed away in that month. Size and transparency of each point reflects the number of decedents in each bin. The red dashed line is the cutoff (reform announcement on March 13$^{\text{th}}$, 2015). The regression lines are based on a linear fit before and after the reform announcement.

Figure 3 clearly indicates a break in levels and trends before and after the reform announcement. While before March 2015, the share of covered decedents was declining, reaching a minimum of around 10% of decedents with cover. After the reform, the level was significantly higher, between 15% and 20%. After the announcement until the end of 2019, the share of covered decedents declines again, though at a much slower rate than before the reform announcement.

Next, Table 3 examines the change in insurance participation and net wealth before and after the announcement of the 2016 tax reform. The table reports average marginal effects from two different probit models of insurance take-up and the treatment effect from a linear model with net wealth as the outcome variable, measured in thousands of Euros. We recover the estimates from our regression discontinuity design (RDD), with automatic bandwidth selection and a triangular Kernel.

Table 3: ATE Baseline Estimates (RDD)

	RDRobust FI (participation)	*RDRobust* LI (participation)	*RDRobust* Net Wealth
	(1)	(2)	(3)
Treatment at cutoff	0.319**	−0.144	112.826
	(0.131)	(0.175)	(84.407)
N	2962	2962	2962
Effective N	297	262	554
Kernel	Triangular	Triangular	Triangular

Note: Column entries are average marginal effects from probit models in columns (1) and (2) and a linear effect in column (3) measured in thousands of Euros. FI (LI) is final expense (other life) insurance. Net wealth refers to net wealth before funeral costs, excluding housing wealth and vehicles. Robust (HC0) standard errors in parentheses; regressions weighted for stratified sampling. Linear smooth on both sides of the cutoff. All models feature year-of-death fixed effects and court district fixed effects. Individuals aged 80 and older at the time of the reform announcement are excluded from the sample.
*p<0.1; **p<0.05; ***p<0.01

The results suggest that FI take-up increased significantly around the reform announcement. The average marginal effect is 0.32. This suggests that among individuals who pass away shortly after the reform announcement, the probability of participation in funeral insurance is by 30 percentage points higher compared to individuals who pass away before the cutoff. The estimate is statistically significant at the five percent level. For other types of life insurance, we do not find an effect of comparable magnitude. The estimate for participation in life insurance other than FI is negative. Moreover, coefficient is estimated imprecisely, and the standard error is larger than the estimate. Finally, column (3) in Table 3 studies net wealth at death before and after the reform.

The estimate suggests that net wealth was lower on the treated side of the cutoff, than among untreated individuals. The estimate is statistically significant at the 10%-level, suggesting that it is less precise, compared to the estimate in column (1). Overall, the non-positive estimate of net wealth implies that even though participation in funeral insurance may have increased, this did come at the expense of other types of savings, at best.

In Section F, we report additional robustness checks for the main specification in Table 3. We re-estimate the all models without weights, and replicate the analysis while including an extensive set of control variables. The results remain qualitatively unchanged: the treatment effect is positive and statistically significant for FI, but not the other outcome variables.

The reform should affect primarily individuals who live on lower incomes due to the diminishing deductibility of insurance contributions above an annual income threshold of €36,000. Therefore, the estimates in Table 3 are likely to mask substantial heterogeneity along the distribution of income. To capture this heterogeneity, Table 4 regresses participation in different insurance types, as well as net wealth on an indicator that distinguishes individuals who pass away before and after the reform announcement, interacted with our measure of income at death.[12] Essentially, the estimation corresponds to an RDD with maximum bandwidth and a uniform kernel. Each model features control variables and fixed effects. We focus on the coefficient estimates identifying the difference in average participation before and after the reform announcement, and the interactions of this coefficient with income.

In column (1) of Table 4, we report the estimate of the difference in mean FI par-

[12]This is only available for a subset of the sample, such that the number of observations drops significantly upon introducing retirement income into the analysis.

Table 4: Income Heterogeneity Estimates

	Dependent variable:		
	probit FI (participation)	*probit* LI (participation)	*OLS* Net wealth
	(1)	(2)	(3)
Post-reform	1.136^{***}	0.482	-521.547
	(0.388)	(0.304)	(397.602)
Post-reform × Income (log)	-0.162^{***}	-0.078^{*}	83.088
	(0.052)	(0.041)	(55.453)
Observations	1,055	1,055	1,055

Note: Column entries are average marginal effects from probit models in columns (1) and (2) and a linear effect in column (3) measured in thousands of Euros. FI (LI) is final expense (other life) insurance. Net wealth refers to net wealth before funeral costs, excluding housing wealth and vehicles. Robust (HC0) standard errors in parentheses; regressions weighted to take into account stratified sampling. Each specification features control variables: age, age (squared), marital status, gender, receipt of long term care cash transfer, intestacy, inter-vivos gifting, and proceeding duration. All models feature year-of-death fixed effects and court district fixed effects. Fixed effects and coefficients on control variables dropped. Individuals aged 80 and older at the time of the reform announcement are excluded from the sample.
*p<0.1; **p<0.05; ***p<0.01

ticipation before and after the reform announcement, controlling for a wide range of covariates. At zero income, the average participation after the announcement was more than twice the participation before the announcement. The estimate is highly significant in statistical terms. In the second row of column (1), we report the estimate of the interaction effect between the post-reform indicator variable and the income coefficient. The estimate suggests that those passing away with higher incomes respond significantly less to the tax reform.

The second column reports the results for the same specification as in column (1) of Table 4, with the exception that the dependent variable is participation in other types of life insurance. Paralleling the estimates in the first column, the coefficient on the effect of LI at zero income is positive, while the coefficient on the interaction effect is negative. However, both effects are imprecisely estimated. The interaction effect is statistically significant at the 10% level.

Finally, column (3) in Table 4 refers to a specification with net wealth at death as the dependent variable. Individuals with low income at death have on average lower wealth when they pass away after the reform announcement, while the opposite effect prevails among those who have higher incomes. However, for neither group, the difference between net wealth before and after the reform announcement is statistically significant.

5 Conclusion and policy implications

This paper examines wealth at death and life-insurance payouts. Using newly digitized probate records from Vienna, we study the availability of wealth and insurance to weather financial shocks towards the end of life. The paper characterizes who holds insurance policies at the end of life and how these holdings vary with terminal wealth and demographics. We then study a tax reform that temporarily increased the relative attractiveness of life insurance.

Three findings emerge. First, a significant minority of decedents has low levels of wealth. As a result, survivors and local governments are liable to cover their final expenses. Insurance payouts in the form of death benefits perform an important buffer role and help families to weather cover final expenses. Second, at low levels of terminal wealth, death benefits primarily insure funeral expenses rather than support survivors' consumption. Other life-insurance products are more closely associated with the presence of survivors but contribute little to terminal wealth across the distribution. Third, take-up of final-expense insurance is responsive to tax incentives, whereas other life insurance and net wealth at death are much less so.

From a policy perspective, governments might have several reasons to seek a reduction in the number of individuals with low levels of wealth in late life. While policymakers face a variety of objectives, many of which vary across countries and over time, it may be desirable to promote wealth creation especially at the bottom of the distribution to reduce economic insecurity and material deprivation. For example, in the European Pillar of Social Rights Action Plan, the European Union commits to reducing number of people at risk of poverty or social exclusion, which can be accomplished through a reduction in material deprivation. Another goal could be to reduce public expenditure on certain transfers such as means tested health benefits or financial support with final expenses. Governments can make progress towards both goals by raising the levels of financial buffers available to individuals in late life to meet high out-of-pocket expenditure needs.

In addition to providing estimates of the prevalence of insufficient financial buffers in late life, this paper informs the choice of policy instruments to address this problem.

While many governments incentivize life insurance as a key pillar to promote financial self-sufficiency among the elderly, the Austrian experience suggests a more cautionary tale. On the one hand, we show that death benefits perform an important role to meet expenditure needs at this point in the life cycle. On the other hand, tax incentivized take-up of life insurance contributes little to the building up of financial buffers. When people use insurance more heavily as a consequence of tax policy, they rely less on other forms of wealth. As a result, their wealth holdings remain more or less unchanged.

As a result, our analysis suggests that tax incentives for insurance might not be policy of choice to reduce the number of elderly individuals who do not have any financial buffers to weather expenditure shocks. An alternative approach could be based on behavioral strategies to increasing financial well-being in old age such as using automatic enrollment in savings plans (Benartzi and Thaler 2013). Moreover, earnings during working life are a strong predictor of low economic security at late stages in the life cycle (Poterba, Venti, and Wise 2018). Removing preferential tax treatment of financial instruments such as the insurance policies discussed in this article and instead reducing the tax burden on individuals and families with low earnings could be more effective. We leave a more rigorous comparison of these policy alternatives for future research.

There are some important qualifications to our findings. First, while our data captures final expense insurance well, the data on life insurances may not have the same quality. It is possible under certain circumstances to circumvent the probate process with specific types of life insurances. While we have data on some of these policies, it is likely that we do not cover all. Second, we trade-off depth and richness of the data against the number of observations in our sample. While this paper benefits from granular information on funeral insurance and household portfolios for example, the treatment effect estimates that we obtain as measures of tax responsiveness are relatively imprecise. Despite these shortcomings, our results provide interesting insights into insurance as a component of household portfolios in late life and tax policy.

References

Alvaredo, Facundo, Anthony B Atkinson, and Salvatore Morelli (2018). "Top wealth shares in the UK over more than a century". In: *Journal of Public Economics* 162. Publisher: Elsevier, pp. 26–47.

Banerjee, Abhijit et al. (2024). "Social protection in the developing world". In: *Journal of Economic Literature* 62.4, pp. 1349–1421.

Benartzi, Shlomo and Richard H Thaler (2013). "Behavioral economics and the retirement savings crisis". In: *Science* 339.6124, pp. 1152–1153.

Berg, Erlend (2018). "Funeral insurance: an inter-generational commitment device?" In: *Journal of African Economies* 27.3, pp. 321–346.

Berman, Yonatan and Salvatore Morelli (Mar. 2021). *On the Distribution of Estates and the Distribution of Wealth: Evidence from the Dead*. en. Tech. rep. w28546. Cambridge, MA: National Bureau of Economic Research, w28546.

Bernheim, B. Douglas (Oct. 1991). "How Strong Are Bequest Motives? Evidence Based on Estimates of the Demand for Life Insurance and Annuities". en. In: *Journal of Political Economy* 99.5, pp. 899–927. ISSN: 0022-3808, 1537-534X.

Brandolini, Andrea, Silvia Magri, and Timothy M Smeeding (2010). "Asset-based measurement of poverty". In: *Journal of Policy Analysis and Management* 29.2, pp. 267–284.

Brunetti, Michael J (2006). "The estate tax and the demise of the family business". In: *Journal of Public Economics* 90.10-11, pp. 1975–1993.

Bulmahn, T. (Mar. 2003). "Zur Entwicklung der privaten Altersvorsorge in Deutschland". In: *Kölner Zeitschrift für Soziologie und Sozialpsychologie* 55.1, pp. 29–54.

Cawley, John and Tomas Philipson (1999). “An empirical examination of information barriers to trade in insurance”. In: *American Economic Review* 89.4, pp. 827–846.

Disslbacher, Franziska et al. (2023). “On top of the top: A generalized approach to the estimation of wealth distributions”. In: *INEQ Working Paper* 22.

Erixson, Oscar and Sebastian Escobar (May 2020). “Deathbed tax planning”. en. In: *Journal of Public Economics* 185, p. 104170. ISSN: 00472727.

Escobar, Sebastian, Henry Ohlsson, and Håkan Selin (Apr. 2023). “Giving to the children or the taxman?” en. In: *European Economic Review* 153, p. 104382. ISSN: 00142921.

Gerba, Eddie and Waltraud Schelkle (2013). “The finance-welfare state nexus”. In.

Gilbert, Neil (2009). “European measures of poverty and” social exclusion”: Material deprivation, consumption, and life satisfaction”. In: *Journal of Policy Analysis and Management* 28.4, pp. 738–744.

Glogowsky, Ulrich (Jan. 2021). “Behavioral responses to inheritance and gift taxation: Evidence from Germany”. en. In: *Journal of Public Economics* 193, p. 104309. ISSN: 00472727.

Gomes, Francisco, Michael Haliassos, and Tarun Ramadorai (Sept. 2021). “Household Finance”. en. In: *Journal of Economic Literature* 59.3, pp. 919–1000. ISSN: 0022-0515.

Gordon, Robert, David Joulfaian, and James Poterba (2016). “Revenue and Incentive Effects of Basis Step-Up at Death: Lessons from the 2010 “Voluntary” Estate Tax Regime”. In: *American Economic Review* 106.5, pp. 662–667.

Gornick, Janet C, Eva Sierminska, and Timothy M Smeeding (2009). “The income and wealth packages of older women in cross-national perspective”. In: *Journals* *of Gerontology Series B: Psychological Sciences and Social Sciences* 64.3, pp. 402–414.

Gropper, Michael J and Camelia M Kuhnen (2025). “Wealth and insurance choices: Evidence from US households”. In: *The Journal of Finance* 80.2, pp. 1127–1170.

Guio, Anne-Catherine et al. (2016). “Improving the measurement of material deprivation at the European Union level”. In: *Journal of European social policy* 26.3, pp. 219–333.

Hacker, Jacob S et al. (2018). “Economic security”. In: *For good measure: Advancing research on well-being metrics beyond GDP*, pp. 203–240.

Hadziabdic, Sinisa and Sebastian Kohl (2022). “Private spanner in public works? The corrosive effects of private insurance on public life”. In: *The British Journal of Sociology* 73.4, pp. 799–821.

Harris, Timothy F and Aaron Yelowitz (2018). “Life insurance holdings and well-being of surviving spouses”. In: *Contemporary Economic Policy* 36.3, pp. 526–538.

Haveman, Robert and Edward N Wolff (2004). “The concept and measurement of asset poverty: Levels, trends and composition for the US, 1983–2001”. In: *The Journal of Economic Inequality* 2.2, pp. 145–169.

Hecht, Carolin and Katja Hanewald (2012). “Who responds to tax reforms? Evidence from the life insurance market”. In: *The Geneva Papers on Risk and Insurance-Issues and Practice* 37.1, pp. 5–26.

Heckman, James J (1979). “Sample selection bias as a specification error”. In: *Econometrica: Journal of the econometric society*, pp. 153–161.

Hong, Jay H and José-Víctor Ríos-Rull (2012). “Life insurance and household consumption”. In: *American Economic Review* 102.7, pp. 3701–3730.

Horn, Alexander and Sebastian Kohl (2024). "Beyond trade-offs: Exploring the changing interplay of public and private welfare provision in old age and health in the historical long-run". In: *Journal of European Social Policy* 34.4, pp. 373–388.

Hurd, Michael D and David A Wise (1989). "The Wealth and Poverty of Widows: Assets before and after the Husband's Death". In: *The economics of aging*. University of Chicago Press, pp. 177–200.

Inkmann, Joachim and Alexander Michaelides (2012). "Can the Life Insurance Market Provide Evidence for a Bequest Motive?" en. In: *Journal of Risk and Insurance* 79.3, pp. 671–695. ISSN: 0022-4367, 1539-6975.

Jappelli, Tullio and Luigi Pistaferri (2003). "Tax incentives and the demand for life insurance: evidence from Italy". In: *Journal of Public Economics* 87.7-8, pp. 1779–1799.

Johannesen, Niels, Johan Sæverud, and Emmanuel Saez (2024). *Taxing the Wealth of the Poor: Evidence from the Danish Old-Age Support Asset Test*. Tech. rep. National Bureau of Economic Research.

Kopczuk, Wojciech (Nov. 2007). "Bequest and Tax Planning: Evidence from Estate Tax Returns". en. In: *The Quarterly Journal of Economics* 122.4, pp. 1801–1854. ISSN: 0033-5533, 1531-4650.

— (Mar. 2016). *U.S. Capital Gains and Estate Taxation: A Status Report and Directions for a Reform*. CEPR Discussion Paper DP11208. Available at SSRN: https://ssrn.com/abstract=2766536. Centre for Economic Policy Research.

Kuypers, Sarah et al. (2022). "Lockdown, Earnings Losses and Household Asset Buffers in Europe 1". In: *Review of Income and Wealth* 68.2, pp. 428–470.

Lin, Yijia and Martin F Grace (2007). “Household life cycle protection: Life insurance holdings, financial vulnerability, and portfolio implications”. In: *Journal of Risk and Insurance* 74.1, pp. 141–173.

Marlier, Eric et al. (2007). “EU indicators for poverty and social exclusion”. In: *The EU and social inclusion: Facing the challenges*. 1st ed. Bristol University Press, pp. 143–196.

Menchik, Paul and Martin David (1983). “Income Distribution, Lifetime Savings, and Bequests”. In: *The American Economic Review* 73.4.

Modigliani, Franco and Richard Brumberg (1954). “Utility analysis and the consumption function: An interpretation of cross-section data”. In: *Franco Modigliani* 1.1, pp. 388–436.

Neumark, David and Elizabeth Powers (May 1998). “The effect of means-tested income support for the elderly on pre-retirement saving: evidence from the SSI program in the U.S.” en. In: *Journal of Public Economics* 68.2, pp. 181–206. ISSN: 00472727.

OECD (2005). *Financial Market Trends: Ageing and Pension System Reform: Implications for Financial Markets and Economic Policies*. Paris: OECD Publishing.

— (2023). *Annual Survey on Financial Incentives for Retirement Savings: OECD Country Profiles 2023*. Tech. rep. Paris: OECD Publishing.

— (2024). *Global Insurance Market Trends 2024*. Paris: OECD Publishing.

Palier, Bruno (2007). “Tracking the evolution of a single instrument can reveal profound changes: the case of funded pensions in France”. In: *Governance* 20.1, pp. 85–107.

Poterba, James, Steven Venti, and David A Wise (June 2018). “Longitudinal determinants of end-of-life wealth inequality”. en. In: *Journal of Public Economics* 162, pp. 78–88. ISSN: 00472727.

Poterba, James and Scott J Weisbenner (2003). "Inter-asset differences in effective estate-tax burdens". In: *American Economic Review* 93.2, pp. 360–365.

Rapp, Severin and Stefan Humer (2023). "Wealth and Welfare: Do Private and Public Safety Nets Compensate for Asset Poverty?" In: *Social Inclusion* 11.1, pp. 176–186.

Sauter, Nicolas and Joachim Winter (2010). "Do investors respond to tax reform? Evidence from a natural experiment in Germany". In: *Economics Letters* 108.2, pp. 193–196.

Slemrod, Joel and Wojciech Kopczuk (2000). *The impact of the estate tax on the wealth accumulation and avoidance behavior of donors*.

Suari-Andreu, Eduard et al. (July 2024). "Giving with a warm hand: evidence on estate planning and inter-vivos transfers". en. In: *Economic Policy* 39.119, pp. 655–700. ISSN: 0266-4658, 1468-0327.

Tomes, Nigel (1981). "The Family, Inheritance, and the Intergenerational Transmission of Inequality". In: *Journal of Political Economy*. 928-958 89.5.

Valentine, Christine and Kate Woodthorpe (2014). "From the cradle to the grave: Funeral welfare from an international perspective". In: *Social Policy & Administration* 48.5, pp. 515–536.

Weir, David R. and Robert J. Willis (2000). "Prospects for Widow Poverty". In: *Forecasting Retirement Needs and Retirement Wealth*. Ed. by Olivia S. Mitchell, P. Brett Hammond, and Anna M. Rappaport. Pension Research Council Publications. Philadelphia: University of Pennsylvania Press. Chap. 7, pp. 208–234. ISBN: 0-8122-3529-0.

Wellschmied, Felix (2021). "The welfare effects of asset mean-testing income support". en. In: *Quantitative Economics* 12.1, pp. 217–249. ISSN: 1759-7323.

Woodthorpe, Kate, Hannah Rumble, and Christine Valentine (2013). “Putting ‘the grave’ into social policy: state support for funerals in contemporary UK society”. In: *Journal of social policy* 42.3, pp. 605–622.

Online Appendix for

"Life Insurance and Death Benefits: What Shapes Participation, and Does It Matter for Inequality?"

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A Data Appendix

A.1 Sampling

The stratified sampling strategy is designed to ascertain that the data better covers the wealth distribution at its extremes. Extreme values are often not included in purely random samples, because in a population with very few extreme values the probability to draw outliers is small. Research on the distribution of wealth uses oversampling in attempts to make wealth surveys more representative. Not at least due to the low probability of drawing a billionaire in a random sample of the population, for example, survey institutes make explicit efforts to include such individuals in the survey by capitalizing on tax data or other external data sources (Vermeulen 2018; Kennickell 2008). To obtain representative results, the population weights are adjusted accordingly, such that the cases with particularly many procedural steps are scaled back to their actual proportion in the population.

Table 5 shows three regression models, where the net probate wealth is regressed on a dummy variable that identifies cases that enter the sample because of the oversampling strategy. In the first column, we report the bivariate correlation, and the second column refers to correlations while controlling for postal code of the last place of residence. The third column refers to a quantile regression on the median value of net wealth. Qualitatively, all models show that the number of procedural steps is positively related to probate wealth. On average, in the first two models, individuals that enter the sample through oversampling leave more than € 1 mio. to their heirs. The final column suggests that the median net wealth is also elevated in the oversampled group.

Table 5: 'Oversampling and mean probate wealth

	Dependent variable:		
	Net wealth		
	LM: no controls	LM: w. controls	Median
	(1)	(2)	(3)
In oversampling	1,092,671.000***	1,110,173.000***	64,260.170***
	(141,257.500)	(150,099.300)	(20,580.460)
Observations	4,629	4,486	4,629
Adjusted R^2	0.012	0.014	0.012
Note:			*p<0.1; **p<0.05; ***p<0.01

A.2 Descriptive statistics

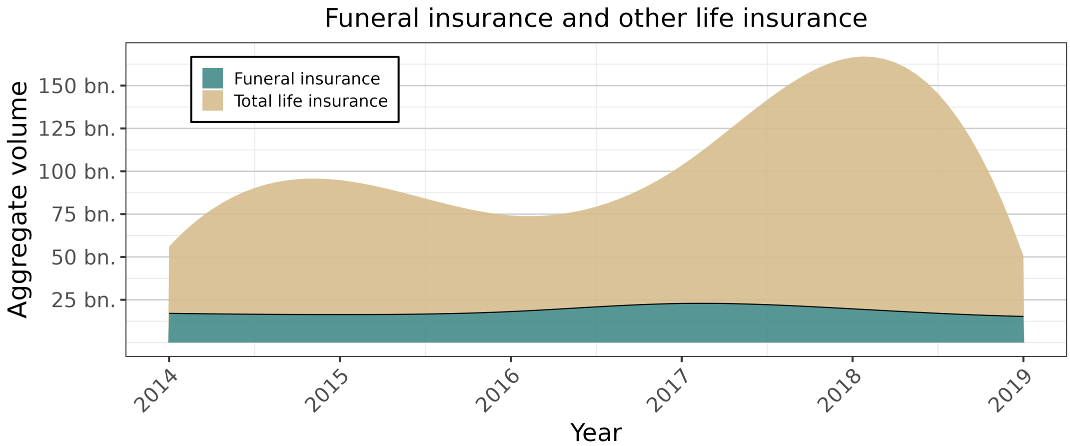
In many other data sources funeral insurances are part of the life insurance aggregate. Figure 4 disentangles these types of insurance. It plots the total volume of life insurance payouts other than final expense insurance policies along the volume of final expense insurance payouts. The estimates result from a scaling exercise with two steps. First, we compute the sum of life and final expense insurance payouts at death at the national level.[13] Then, we use the inverse mortality rate by age groups to obtain a measure of the aggregate among the living. The second step follows the logic of the mortality multiplier method to estimate wealth among the living from data on terminal wealth of decedents.

Figure 4 shows that funeral insurance ranges below 20 billion over time. The estimates of life insurance are much more volatile. In 2014 and 2016, for example, the aggregate value of life insurance other than final expense insurance payouts are below € 50 bio. The estimate peaks at almost € 150 bio. in 2018. The volatility is a result of the fact that we sample from the decedent population, which is already a sample of the

[13] Our dataset covers 7.79% of the relevant Austrian reference population. The HFCS suggests that on average, wealth in the Austrian districts that we do not cover exceeds mean wealth in the 10 Viennese districts of this study by more than 40%. We use this information for a back-of-the-envelope-calculation to derive the national volume.

living. Therefore, outliers are likely to result in more volatile estimates. In Appendix A.2, Figure 5 shows that our aggregate estimate of life and final expense insurance oscillates around the aggregate reported by by the Austrian National Bank and the ECB distributional wealth accounts.

Figure 4: Aggregate value of funeral insurance and other life insurance over the sampling period

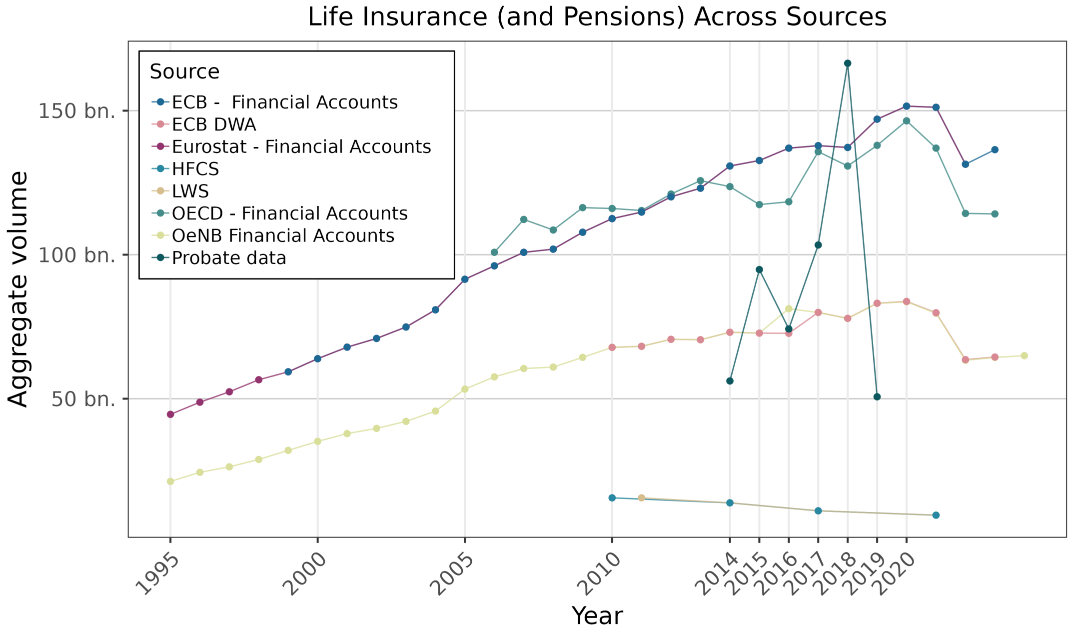


Note: Estimates obtained from weighting observations by the sample weight and the inverse district-specific mortality rate. Wealth outside the court districts in the sample is scaled by 1.4 to reflect higher average wealth outside Vienna (Dabrowski et al. 2020). Smoothed curves are obtained by fitting an FMM cubic spline to the seven yearly observations for each series and stacking the interpolated values.

Figure 5 plots the aggregate value of life insurance implied by the probate data against the volume suggested by other sources, including the financial accounts data and survey data (Morelli et al. 2023). Overall, our data is consistent with the European Central Bank's (ECB) Distributional Wealth accounts estimate and those by the Austrian National bank. They are below the ECB financial accounts, and those compiled by Eurostat and the OECD. This is due to the fact that the latter three aggregates also include other pension products.

Table 6 shows the covariate balance across individuals who pass away with a final

Figure 5: Source comparison on aggregates of life insurance and voluntary pensions.



Note: Probate series estimates obtained from weighting observations by the sample weight and the inverse district-specific mortality rate. Wealth outside the court districts in the sample is scaled by 1.4 to reflect higher average wealth outside Vienna (Dabrowski et al. 2020). Other sources are taken from Morelli et al. (2023).

expense insurance and those who are not covered. We report means for each variable in the third and fourth column, as well as the difference in means in the fifth column. The final column reports the p-values for a two-sided t-test of the differences in means. The table complements the regression analysis in Table 2. However, Table 6 only considers bivariate associations.

Table 7 reveals several important facts. First, while in almost all groups, the most people are able to weather a certain expenditure shock, there is always a non-trivial minority that does not. For example, only 63% of decedents that die as members of private households have sufficient gross wealth to cover their final expenses, while 37%

Table 6: Characteristics of funeral insurance policy holders

Section	Variable	No policy	Policy holder	Difference	p-value
Age	Share 0 to 30 years	0.02	0.01	0.02	0.00
	Share 30 to 60 years	0.12	0.04	0.08	0.00
	Share 60 to 70 years	0.13	0.08	0.05	0.00
	Share 70 to 80 years	0.24	0.24	0.01	0.95
	Share 80 to 90 years	0.25	0.27	-0.02	0.23
	Share 90 to 100 years	0.22	0.34	-0.12	0.00
	Share 100 to 120 years	0.01	0.02	0.00	0.54
	Missing	0.00	0.00	0.00	0.32
Gender	Share male	0.49	0.36	0.13	0.00
	Share female	0.51	0.64	-0.13	0.00
Marital status	Share single	0.36	0.26	0.09	0.00
	Share divorced	0.34	0.50	-0.16	0.00
	Share widowed	0.00	0.00	0.00	0.00
	Share married	0.16	0.14	0.02	0.08
	Share other	0.12	0.09	0.04	0.03
	Missing	0.01	0.01	0.01	0.06
Long-term care need	Share with needs	0.23	0.17	0.06	0.00
	Share without needs	0.32	0.41	-0.09	0.00
	Missing	0.45	0.42	0.03	0.06
Testament	Share with testament	0.68	0.59	0.09	0.00
	Share intestate	0.32	0.41	-0.09	0.00
Made gift	Share made gift	0.95	0.95	0.00	0.85
	Share no gift	0.05	0.05	0.00	0.85
Other life insurance	Share with other LI	0.90	0.72	0.18	0.00
	Share without other LI	0.10	0.28	-0.18	0.00
Duration of probate process	Proceeding duration	477.18	272.03	205.15	0.10
Weight	Weight	9.41	9.56	-0.15	0.12

Note: Means for key sociodemographics for decedents with and without final expense insurance. Difference is the mean among those without final expense insurance minus the mean among policy holders. P-values from Welch t-test. Weights account for stratified sampling.

do not have the means to do so.[14] Second, if all death benefits were eliminated, the share of individuals who are unable to fund their final expenses climbs by three percentage points. Third, the share of individuals with sufficient buffers at death is lower in care

[14]Gross wealth is a meaningful indicator for this type of analysis, because final expenses are priority debt, while all other claims are junior debt

homes. Compared to the 63% of private household members with sufficient gross wealth to cover final expenses, it is only 58% in care homes. Interestingly, death benefits are somewhat more important among care home residents.

Table 7 reports the share of individuals whose wealth exceeds a given expenditure shock, distinguishing between individuals in private households and those in care homes. The upper panel refers to gross and net wealth including death benefits. The lower panel excludes death benefits from the wealth definition. We report both the share of individuals whose wealth exceeds their own funeral cost and the share of individuals who cannot afford final expenses of € 5,000 (which is slightly below the average cost of a funeral).

Table 7: Affordability of funeral costs by care-home status

	Not in care home		Care home	
	Gross	Net	Gross	Net
Population share	79%		21%	
Wealth at death				
Afford actual funeral	63%	59%	58%	54%
Afford €5,000 funeral	61%	59%	56%	53%
Wealth at death net of death benefit				
Afford actual funeral	60%	57%	52%	50%
Afford €5,000 funeral	58%	57%	50%	49%

Note:
Population share is centered across the two columns for each group. The affordability measures reflect the share of decedents who can afford a given expenditure shock. "actual funeral" uses the observed cost of each decedent's funeral. We compare this to the cost of a hypothethical funeral worth €5,000. Weights are used to take into account stratified sampling.

B Final expenses and insurance

Table 8 offers an overview of the funeral insurance policies offered in Austria in 2017 (Verein für Konsumenteninformation 2017). The table lists the annual premium for a typical contract with for a face value of €5,000.00 and 10,000.00. It also indicates whether a medical questionnaire is required.

Table 8: Final Expense Insurance Comparison (Austria)

Insurer	**Premium/yr (€5,000)**	**Premium/yr (€10,000)**	**Health questions**	**Max. age**
Donau	656.00	1,189.40	No	70-75
ERGO	666.42	1,211.29	No	70-75
Generali	688.58	1,361.58	Yes	80
Helvetia	653.57	1,046.08	Yes	85
Merkur	708.14	1,320.59	No	85
Oberösterreichische	609.65	1,130.04	No	70-75
UNIQA	681.65	1,182.00	No	70-75
Wiener Verein	656.00	1,189.40	Yes	80
Wüstenrot	736.75	1,380.82	Yes	70-75
Zurich	691.10	1,259.95	No	70-75

Notes: Figures are annual premia for stated benefit sums; pricing reference is September 2017. "Health questions" indicates whether medical questions are asked at application. Maximum age refers to the age threshold up to which insurers offer contracts. In some cases, individuals close to the threshold have to pay all premia in one lump sum.
Source: Verein für Konsumenteninformation (2017)

C Theory

C.1 Setup

We start from a simple setting where agents can either choose to buy funeral insurance or to self-insure. In this world, funeral insurance serves two purposes. In both cases, the final expense insurance serves as a commitment device. On the one hand, survivors rather than decedents choose the level of funeral expenditure. As the payout of final expense insurance is earmarked, decedents can set their preferred level of funeral expenditure by choosing the level of funeral insurance accordingly. This is particularly attractive if survivors would choose a different level of funeral spending than the decedent (in order to increase their inheritance, for example). On the other hand, we allow for individuals with time inconsistent preferences. Decedents may use funeral insurance as a tool to ensure that they do not run down their wealth to quickly without covering their funeral. If the younger self knows that in the future, they will prefer to consume rather than set money aside for a funeral, they can choose insurance to commit themselves to saving.

In the latter case, a simple whole life insurance with low face value might perform a similar role to the funeral insurance. Therefore, the paper discusses two extensions in the Appendix where individuals can also choose from other life insurance products and show under what conditions funeral insurance are still preferable to life insurance.

Consider a simple model with three periods $t \in \{0, 1, 2\}$. In period 0, the agent simply chooses the optimal level of funeral insurance A. For simplicity, we assume that the agent pays the entire premium as a single premium in this period - an option that most insurers offer.

As a result, at the beginning of period 1, the agent has a level of liquid wealth S_1 given by $W - pA$, where W is initial wealth, and p is the price of the insurance product.[15] At $t = 1$, the agent either dies with probability $1 - s$ or survives with probability s and dies in $t = 2$ with certainty. Survivors spend $A + T_t$ on the funeral, where T_t is a top-up they choose. $W - A - F_t$ is left as a bequest B_t to the survivor.

Next, assume that the agent maximizes a utility function. The agent has CRRA util-

[15] This can be thought of as the profit of the insurance company, or an administrative charge or tax.

ity from consumption, the funeral expenditure and a bequest, which is a luxury good.[16] Then, the agent's utility is given by:

$$
\begin{aligned}
U(W) =& (1-s)\left[\frac{\mu(F_1+A)^{1-\gamma}}{1-\gamma}+\frac{\phi_1}{1-\gamma}\left(\frac{B_1}{\phi_1}+\phi_2\right)^{1-\gamma}\right]+ \\
& s\left(\frac{C^{1-\gamma}}{1-\gamma}+\frac{\beta_t}{1+\rho}\left[\frac{\mu(F_2+A)^{1-\gamma}}{1-\gamma}+\frac{\phi_1}{1-\gamma}\left(\frac{B_2}{\phi_1}+\phi_2\right)^{1-\gamma}\right]\right)
\end{aligned}
\tag{1}
$$

In contrast, the survivor (d) can allocate the liquid wealth they receive between funeral expenditure and their inheritance. Their preferences are given by:

$$
U_d(W,A) = \left[\frac{\mu(F_t+A)^{1-\gamma}}{1-\gamma}+\frac{\phi_1}{1-\gamma}\left(\frac{B_t}{\phi_1}+\phi_2\right)^{1-\gamma}\right] \tag{2}
$$

μ is a parameter that determines how much utility weight the funeral has. It is unity for the decedent. However, we use the parameter to open the possibility that the weight placed on funeral versus bequests differs between the agent and the survivor with otherwise similar preferences ($\mu \neq \mu_d$), giving rise to an interpersonal conflict of interest. γ, μ, ϕ_1 and ϕ_2 are utility function parameters. To ensure that the agent treats B as luxury goods, $\phi_2 > 0$. ρ is a standard time discount factor. We use β_t to implement time-inconsistency that gives rise to demand for interpersonal commitment tools. In

[16]The functional form of the bequest motive is taken from Kvaerner (2023), Lockwood (2018) and Ameriks et al. (2011). Each parameter of the bequest motive has an interpretation. ϕ_2 is a minimum floor. Individuals with wealth below the floor do not leave a bequest. ϕ_1 is the strength of the bequest motive. The "warm-glow" specification (Andreoni 1989) in this paper is a reduced form for the altruistic bequest motive, where ϕ_2 is a function of the present discounted value of the combined labor income across all future generations (Ameriks et al. 2011; Abel and Warshawsky 1988). For simplicity, we assume that the risk aversion parameter γ for bequests coincides with the risk aversion parameter for consumption (Kvaerner 2023; De Nardi and Yang 2014; De Nardi 2004; Ameriks et al. 2011).

$t = 0$, $\beta = 1$, while in $t = 1$, $0 < \beta \leq 1$.

Solution Given A, if death occurs at $t = 1$, the survivor chooses (T_1, B_1) to allocate S_1. Otherwise, the decedent chooses $S_2 = S_1 - C$, before the survivor decides on allocating S_2 between T_2 and B_2.

Let $V(S, A; \mu_c)$ be the value of the (T, B) problem at liquid wealth S and coverage A with survivor weight μ_c. The $t = 1$ consumption choice satisfies the Euler equation

$$C^{-\gamma} = \frac{\beta}{1+\rho} M_S(S_2, A; \mu_c), \qquad M_S(S, A; \mu_c) \equiv \frac{\partial V}{\partial S}(S, A; \mu_c) > 0. \tag{3}$$

Insurance choice The decedent chooses to maximize (1), anticipating (3) and the (T, B) rules in each branch. Earmarking (interpersonal commitment) operates through μ vs. μ_c; intrapersonal commitment operates through $\beta < 1$, which reduces S_1 available for $t = 1$ consumption and raises the shadow value of S_2.

C.2 Statics

Wealth gradients and participation Several commitment channels can generate an inverted u-shape for the relationship between funeral insurance participation and wealth. For low W coverage is unaffordable; at high W the marginal value of earmarking/-commitment declines (bequests less luxury-constrained, self-control less binding). A minimum policy reinforces this pattern.

Price cuts Regardless of the insurance motive, price cuts (or tax changes) do not change participation for some individuals, but raises it for others. Other implications depend on the insurance motive. If $\beta = 1$ and $\mu_c < \mu$, then a decrease in p weakly increases coverage (A^*). On any wealth range where (T_t, B_t) are interior, expected funeral spending increases. If $\beta < 1$ and $\mu_c = \mu$, then a decrease in p can raise participation and A^* but does not raise funeral spending upon death at $t = 1$. In contrast, with $p > 1$ it weakly reduces T_1^* because total resources $S_1 + A = W - (p-1)A$ fall with A. The survival branch reallocates toward $t = 2$ (lower C, higher S_2), so T_2^* may increase, but the expected change in the funeral spending by survivors is small and can be non-positive for moderate s.

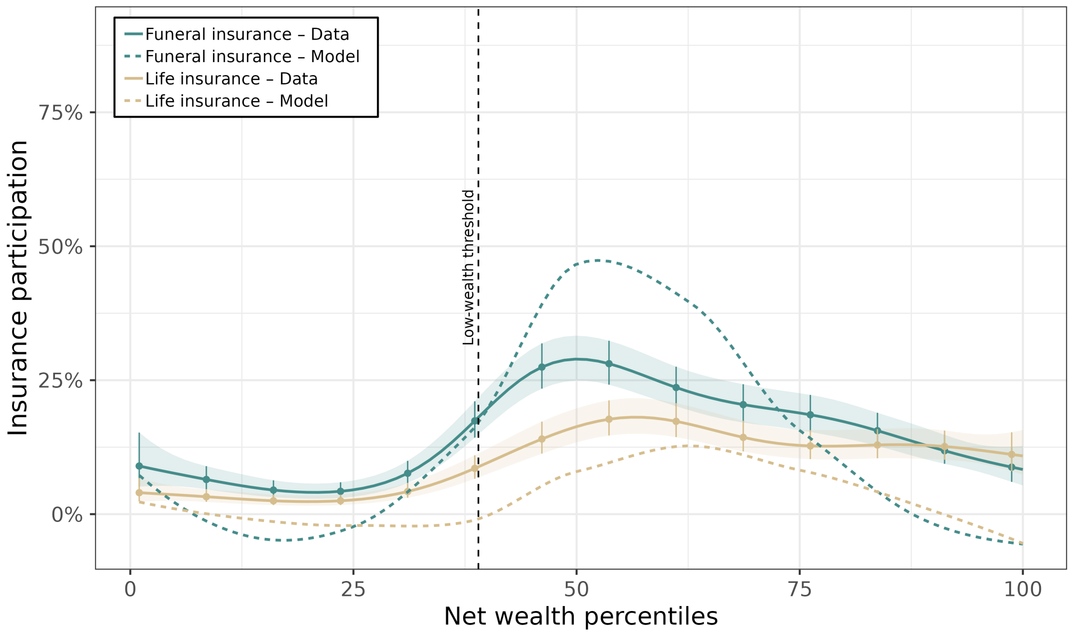
D Model dynamics

Figure 6 shows participation in different types of life insurance along the distribution of wealth at death. It plots a smoothed estimate (locally estimated scatterplot smoothing) of the mean participation in each percentile on the y-axis, and the range of percentiles on the x-axis. The upper panel refers to the data (residualized for age), and the lower panel to the model. Both the data and the model point towards a hump-shape of insurance participation in wealth. The inverted u-shape is much more pronounced for FI.

Figure 7 illustrates how funeral insurance participation evolves along the distribution of wealth. Each panel illustrates what happens upon varying one of the key model parameters while holding the others constant. The first panel examines changes in β, and the second changes in μ_c. The two lower panels examine the changes in participation along the wealth distribution with either β or μ_c fixed at 0.75, while the price p varies.

As baseline parameters, we stipulate $s = 0.5$, $\mu = 1$ and $\rho = 0.05$. For the bequest motive, we take the estimates of Ameriks et al. (2011), and set $\phi_1 = 47.6$ and $\phi_2 = 7,280$.

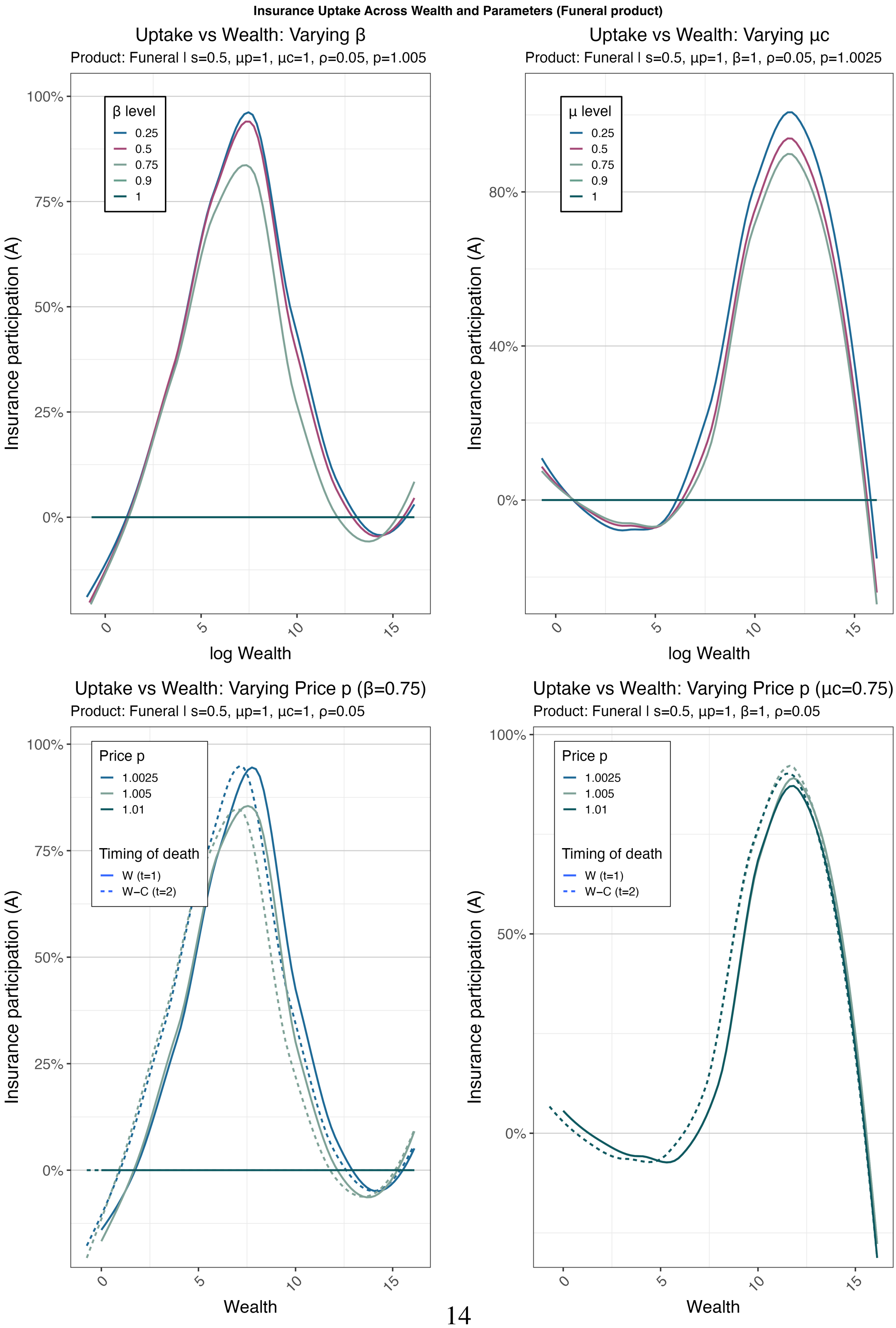
Figure 6: Insurance take-up in the data and the model



Note: The x-axis in each panel refers to the percentiles of the distribution of wealth at death, and the y-axis to participation in insurance. In the left panel, we report participation as measured in our data (across years). The estimates are residualized for age. The standard errors are based on the sampling probability. Weights are used to take into account stratified sampling. In the right panel, we plot the participation rates from our model. In the model wealth values are sampled on a logarithmic grid from 1 to 10,000,000 (100 points), so spacing is even in log space. $\mu = 1$, $\beta = 0.75$,$\rho = 0.05$, $p = 1.005$, $s = 0.99$, and the rest of the parameters is taken from Ameriks et al. (2011) $\gamma = 3$ $\phi_1 = 47.6$ and $\phi_2 = 7,280$. The mean participation in each percentile is estimated with a loess smooth. Low wealth is defined as terminal net wealth below one quarter of the annual income at the poverty line for a single individual.

In each panel, we show a loess-smoothed estimate for the mean participation rate. The upper two panels refer to a cross section of individuals dying in the first period with wealth W and the second period where wealth is $W - C$. In the lower two panels, we separate the different death cohorts.

Figure 7: Model dynamics



E The 2016 Reform

In addition to changing the incentives for personal insurances, the tax reform also affected other elements of the tax system. First, the tax rates for different income brackets changed. Table 9 illustrates the tax schedule before and after the reform. In addition to changes in the tax schedule, another significant components of the reform concerns the transfer of real estate within the family. Before the reform, property wealth passed on from parents to children was valued at the administrative cadastral value for the property transfer tax assessment. Due to the reform, the valuation changed to market values, which significantly increased the tax on real estate transferred across generations within the family.

Table 9: Income Tax Brackets and Rates Before and After the 2016 Reform

Pre-reform (2015)		Post-reform (2016)	
Income (€)	**Rate (%)**	**Income (€)**	**Rate (%)**
0 – 11,000	0	0 – 11,000	0
11,001 – 25,000	36.5	11,001 – 18,000	25
25,001 – 60,000	43.214	18,001 – 31,000	35
60,001 – 150,000	50	31,001 – 60,000	42
150,001 – 1,000,000	50	60,001 – 90,000	48
Over 1,000,000	50	90,001 – 1,000,000	50
		Over 1,000,000	55

Rates shown are nominal statutory tax rates across tax brackets. Source: Schratzenstaller (2015).

Table 10: ATE Estimates Without Weights (RDD)

	Outcome		
	FI (participation)	LI (participation)	Net Wealth
Treatment at cutoff	0.420^{*}	-0.024	-342.476
	(0.236)	(0.145)	(218.934)
N	2962	2962	2962
Kernel	Triangular	Triangular	Triangular
Observations	32	32	32

Note: Column entries are average marginal effects from probit models in columns (1) and (2) and a linear effect in column (3) measured in thousands of Euros. FI (LI) is final expense (other life) insurance. Net wealth refers to net wealth before funeral costs, excluding housing wealth and vehicles. Robust (HC0) standard errors in parentheses; regressions weighted for stratified sampling. Linear smooth on both sides of the cutoff. All models feature year-of-death fixed effects and court district fixed effects. Individuals aged 80 and older at the time of the reform announcement are excluded from the sample.

$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

Table 11: ATE Estimates With Controls (RDD)

	Outcome		
	FI (participation)	LI (participation)	Net Wealth
Treatment at cutoff	0.377^{***}	-0.009	273.770^{***}
	(0.140)	(0.187)	(98.842)
N	2648	2648	2648
Kernel	Triangular	Triangular	Triangular
Observations	32	32	32

Note: Column entries are average marginal effects from probit models in columns (1) and (2) and a linear effect in column (3) measured in thousands of Euros. FI (LI) is final expense (other life) insurance. Net wealth refers to net wealth before funeral costs, excluding housing wealth and vehicles. Robust (HC0) standard errors in parentheses; regressions weighted for stratified sampling. Each specification features control variables: age, age (squared), marital status, gender, receipt of long term care cash transfer, intestacy, inter-vivos gifting. All models feature year-of-death fixed effects and court district fixed effects. Linear smooth on both sides of the cutoff. Individuals aged 80 and older at the time of the reform announcement are excluded from the sample.

$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

F Alternative Treatment Effect Estimates

G Survivor responses to planning strategies

In our final set of results, we examine whether policy-induced variation in final expense planning is related to survivors' allocation of resources between funeral spending and bequests. The upper panel of Table 12 presents the first-stage estimates, where the dependent variable is FI take-up.[17] The lower panel reports the second-stage results, linking insurance take-up to funeral spending. Column (1) estimates a linear specification without covariates beyond fixed effects, while Column (2) adds further covariates. Column (3) drops weights.

Across specifications, the first stage reaffirms the relationship between the reform and the funeral insurance take-up. Across all specifications, the effect of the reform on participation is statistically significant at least at the 5% level. However, the treatment effect in the first stage is largest in magnitude when we introduce additional control variables.

The estimates of the second stage suggest that the reform-induced changes in funeral insurance take-up did not affect funeral spending. Across specifications, the coefficient estimate ranges between approximately a negative € -780.00 and positive € 3,190.00. However, regardless of the choice of the specification, the estimate of the effect of funeral insurance take-up on spending remains statistically insignificant in all three columns.

[17] In contrast to Table 3, in Table 12 the MSE-optimal bandwidth selector takes into account the second stage (Calonico, Cattaneo, and Titiunik 2015). Moreover, the first stage of the fuzzy RD is a linear probability model. As a result, the coefficient interpretation is different between the the results in Tables 12 and the first column in Table 3.

Table 12: Tax reform and Funeral Spending (Fuzzy RDD)

	(A) First stage: Take-up		
	(1)	(2)	(3)
Local Wald first stage	0.169*	0.281**	0.165*
	(0.089)	(0.112)	(0.090)
	(B) Second stage: Spending		
	(1)	(2)	(3)
Local Wald second stage	3,193.465	620.161	−781.655
	(6,272.860)	(3,944.028)	(4,987.809)
Estimation bandwidth (h)	180.4	150.2	192
Eff. Number of Obs.	464	331	494

Note: First and second stage treatment estimates of fuzzy RDD. The dependent variable in the first stage is funeral insurance uptake. The second stage dependent variable is funeral spending in EUR. Robust standard errors in parentheses. Linear smooth on both sides of the cutoff. All models feature year-of-death fixed effects and court district fixed effects. Column (2) has age, age (squared), marital status, gender, receipt of long term care cash transfer, intestacy and inter-vivos gifting as additional control variables. Column (3) drops weights. The first stage is a linear probability model and the second stage an OLS regression, both implemented with the rdrobust R package (Calonico, Cattaneo, and Titiunik 2015). Individuals aged 80 and older at the time of the reform announcement are excluded from the sample.

*p<0.1; **p<0.05; ***p<0.01

References

Abel, Andrew B and Mark Warshawsky (1988). "Specification of the joy of giving: Insights from altruism". In: *The Review of Economics and Statistics* 70.

Ameriks, John et al. (2011). "The joy of giving or assisted living? Using strategic surveys to separate public care aversion from bequest motives". In: *The journal of finance* 66.2, pp. 519–561.

Andreoni, James (1989). "Giving with impure altruism: Applications to charity and Ricardian equivalence". In: *Journal of political Economy* 97.6, pp. 1447–1458.

Calonico, Sebastian, Matias D Cattaneo, and Rocio Titiunik (2015). "rdrobust: An r package for robust nonparametric inference in regression-discontinuity designs". In.

Dabrowski, Cara et al. (2020). *Vermögen in wien: Ungleichheit und Öffentliches eigentum*. Economics of Inequality (INEQ), Wirtschaftsuniversität Wien.

De Nardi, Mariacristina (July 2004). "Wealth Inequality and Intergenerational Links". en. In: *The Review of Economic Studies* 71.3, pp. 743–768. ISSN: 1467-937X, 0034-6527.

De Nardi, Mariacristina and Fang Yang (Nov. 2014). "Bequests and heterogeneity in retirement wealth". en. In: *European Economic Review* 72, pp. 182–196. ISSN: 00142921.

Kennickell, Arthur B (2008). "The Role of Over-sampling of the Wealthy in the Survey of Consumer Finances". In: *Irving Fisher Committee Bulletin* 28.August, pp. 403–08.

Kvaerner, Jens Soerlie (July 2023). "How Large Are Bequest Motives? Estimates Based on Health Shocks". en. In: *The Review of Financial Studies* 36.8. Ed. by Tarun Ramadorai, pp. 3382–3422. ISSN: 0893-9454, 1465-7368.

Lockwood, Lee M. (Sept. 2018). “Incidental Bequests and the Choice to Self-Insure Late-Life Risks”. en. In: *American Economic Review* 108.9, pp. 2513–2550. ISSN: 0002-8282.

Morelli, Salvatore et al. (2023). *The GC Wealth Project Data Warehouse v.1 - Documentation*.

Schratzenstaller, Margit (2015). “Steuerreform 2015/16-Maßnahmen und Gesamteinschätzung”. In: *WIFO-Monatsberichte* 88.5, pp. 371–385.

Verein für Konsumenteninformation (Oct. 25, 2017). *Bestattungskosten: Sparbuch schlägt Versicherung*. Aktualisiert am 22.12.2017. VKI. (Visited on 09/15/2025).

Vermeulen, Philip (2018). “How fat is the top tail of the wealth distribution?” In: *Review of Income and Wealth* 64.2, pp. 357–387.